CSE40467 - Data Science
Milestone 1
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Team Number	4
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Date	5/7/2026

Instructions:
- Maximum length: 20 pages, excluding members' contributions and references.
- Put your responses inside the boxes.
- You can add figures in the boxes.
- Do not alter the font size (11-point), line spacing (1-line), or font typography (Times New Roman).
- References should be added by the end of the document. Use the format that you prefer (e.g., APA, IEEE, ACM, etc.)
- Please do not forget to cite when you make claims or references.

1. Introduction

1.1. Executive Summary. *A good executive summary is a concise and clear overview of a larger document that highlights its key points, findings, and recommendations, aimed at giving readers a quick and comprehensive understanding of its content.*

This project analyzes behavioral data from remote workers to predict burnout risk, identify key drivers of burnout, and uncover underlying employee behavior patterns. Using a dataset of 2,000 observations, we apply a combination of classification, regression, and clustering techniques to develop a comprehensive understanding of burnout in work-from-home environments.

For prediction, we implement three classification models: Logistic Regression, Random Forest, and SVM. We used these to classify employees as either low risk or at risk of burnout. After addressing class imbalance and removing features that introduced target leakage, all models achieved strong performance, with accuracies ranging from 94% to 96%. The SVM model performed best overall, indicating the presence of nonlinear relationships in burnout prediction.

To better understand what drives burnout, we used both OLS and ML regression models. Results show that well-being factors, such as sleep, fatigue, and isolation, explain significantly more variation in burnout than work-related factors alone. This suggests that burnout is not solely driven by workload, but also by behavioral and psychological conditions.

Clustering models were used to identify distinct employee profiles without using outcome labels. Across all methods, three consistent groups emerge: low burnout, moderate burnout, and high burnout. However, relatively low silhouette scores indicate that these groups overlap, suggesting that burnout exists on a spectrum rather than in clearly defined categories.

Overall, this project demonstrates that burnout can be effectively predicted and analyzed using behavioral data. The findings highlight the importance of both work patterns and well-being factors, and suggest that organizations can use such models to proactively identify at-risk employees and design targeted interventions to improve employee health and productivity.

*1.2. **Problem definition**. Explain what this problem is about. Discuss why it should be studied/analyzed.*

This problem focuses on understanding and predicting employee burnout in remote work environments. With the rise of remote work, employees face new challenges such as blurred work-life boundaries, increased screen time, and social isolation, all of which can contribute to burnout. Burnout is a critical issue because it affects employee well-being, productivity, and long-term organizational performance. By analyzing behavioral and well-being data, we aim to predict burnout risk and identify the key factors that contribute to it.

2. Related Work

2.1. Describe related work with this data source.

Burnout and social isolation among remote workers have been widely studied in psychology and human-computer interaction (HCI) research. Prior work finds that behavioral signals such as excessive work hours, after-hours work, reduced sleep, and frequent context switching are strongly associated with fatigue and burnout. Recent large-scale surveys further show that fully remote workers report higher levels of stress and loneliness compared to hybrid or onsite workers, underscoring the growing importance of predictive models in this domain (Gallup, 2025).

In industry settings, several techniques have been developed to handle survey-based or Likert-style variables. One approach is extreme sampling, which focuses on the top and bottom portions of a distribution (e.g., top 30% and bottom 30%) while discarding the middle, thereby reducing ambiguity and strengthening signal clarity. Another common approach is threshold-based binarization, where ordinal responses are converted into binary categories (e.g., low vs. high risk). Additionally, Principal Component Analysis (PCA) is frequently used for dimensionality reduction, allowing researchers to identify the most important underlying factors contributing to burnout. PCA can also support downstream tasks such as clustering and behavioral profiling. For example, individuals may be grouped into profiles such as high workload and high isolation, low workload and high fatigue, or balanced/low-risk categories. Unsupervised learning techniques such as K-Means clustering are often applied to uncover these latent behavioral patterns without requiring labeled outcomes (Data Products, 2025). These approaches enable organizations to better understand employee segments and design targeted interventions.

In addition to prior academic and industry approaches, we examined an existing Kaggle notebook that applies machine learning techniques to the same dataset (Shinde, 2026). Rather than directly predicting burnout risk categories, this study first models burnout score as a continuous variable using a Random Forest regression model. This approach is motivated by the need to avoid label leakage and overly optimistic classification performance. Predicted burnout scores are then converted into risk categories using data-driven thresholds, reflecting real-world decision-making processes where risk levels are inferred from continuous indicators. While this method produces reasonable regression accuracy (RMSE ≈ 6.12), its classification performance is limited, achieving approximately 57.5% accuracy when mapping predicted scores to risk categories. This highlights the challenges of class imbalance and threshold-based conversion, and motivates the use of direct classification approaches in our project.

3. Data description

*3.1. **Data Collection**. Explain how data collection was conducted. If your dataset comes from a website, describe the source and explain how the original researchers potentially obtained the data.*

The dataset used in this analysis was obtained from Kaggle, an online platform that hosts publicly available datasets for research and machine learning applications. According to the dataset description, the data is synthetically generated, meaning it was not collected from real individuals but instead created using simulated patterns designed to resemble realistic work-from-home behaviors. As a result, while the dataset reflects plausible relationships between variables, it may not fully capture the complexity or variability of real-world data.

3.2. ***Data Preprocessing.*** *Explain all the steps that you performed to transform the raw data into a manageable dataset (e.g., merge, impute missing values, standardize, normalize). If your data did not require any data pre-processing, please discuss any potential quality issues of the data (e.g., how representative is the dataset? Who pre-processed the dataset?)*

The dataset required minimal preprocessing due to its synthetic nature. First, the data was checked for missing values and inconsistencies, and the user ID column was dropped as it was not relevant to the analysis. Data types were then verified to ensure all numerical variables were correctly formatted, while categorical variables were encoded for modeling; specifically, one-hot encoding was applied to the weekday column. To ensure comparability across features, numerical variables were standardized using z-score normalization. The dataset was also briefly examined for outliers, though few were present given the synthetic generation process. Additionally, the data was split into training and testing sets using an 80/20 split, where X represents all the input features and Y represents burnout risk. Overall, preprocessing was straightforward, and much of the data appears to have been pre-cleaned by the dataset creator; however, its synthetic nature means it may not fully capture real-world variability or noise. In addition, the original target variable exhibited class imbalance, with fewer observations in the High burnout category. To address this, Medium and High burnout levels were combined into a single “At Risk” class, resulting in an approximately balanced binary classification problem. This transformation simplifies the prediction task while preserving meaningful distinctions between low-risk and at-risk individuals.

3.3. ***Data Documentation.*** *Describe all the features/variables of your dataset. Please describe their type of data, ranges, min, max, or mode.*

The dataset consists of 2,000 observations of remote workers and contains behavioral and well-being variables. There are no missing values. Most variables are numerical, with a small number of categorical and binary features.

Independent Variables (Features)

- work_hours *(float)*
  - Range: 4.73 to 14.23 hours
  - Description: Total hours worked per day
  - Interpretation: Higher values indicate heavier workload
- screen_time_hours *(float)*
  - Range: 2.91 to 13.98 hours
  - Description: Time spent on screens
  - Interpretation: Proxy for digital workload intensity
- meetings_count *(int)*
  - Range: 0 to 17
  - Description: Number of meetings attended per day
  - Interpretation: Higher values may indicate fragmented work
- breaks_taken *(int)*
  - Range: 0 to 11
  - Description: Number of breaks taken

- Interpretation: More breaks may support recovery
- after_hours_work *(binary: 0/1)*
  - Values: 0 = No, 1 = Yes
  - Description: Indicates whether work continues after normal hours
  - Interpretation: Reflects poor work-life boundaries
- app_switches *(int)*
  - Range: 5 to 152
  - Description: Number of application switches
  - Interpretation: Proxy for multitasking and cognitive load
- sleep_hours *(float)*
  - Range: 3.58 to 9.22 hours
  - Description: Hours of sleep per night
  - Interpretation: Lower values indicate poorer recovery
- task_completion *(float)*
  - Range: 40.0 to 100.0 (%)
  - Description: Percentage of tasks completed
  - Interpretation: Measure of productivity
- isolation_index *(int)*
  - Range: 3 to 9
  - Description: Social isolation score
  - Interpretation: Higher values indicate greater isolation
- fatigue_score *(float)*
  - Range: 1.45 to 10.0
  - Description: Level of fatigue
  - Interpretation: Higher values indicate more exhaustion
  - *Note: Removed from classification models due to potential leakage*
- burnout_score *(float)*
  - Range: 6.06 to 82.22
  - Description: Continuous burnout intensity score
  - Interpretation: Higher values indicate greater burnout
  - *Note: Excluded from classification to prevent leakage*
- day_type_Weekend *(binary: 0/1)*
  - Derived from: `day_type`
  - Values: 0 = Weekday, 1 = Weekend

Dependent Variable (Target)

- burnout_risk *(categorical)*
  - Categories:
    - Low: 50.95%
    - Medium: 42.15%
    - High: 6.90%

This distribution indicates class imbalance, particularly with fewer high-risk observations.

To address this, the target variable was transformed into:

- burnout_risk_binary *(binary)*
  - 0 = Low
  - 1 = Medium or High (At Risk)

3.4. ***Variables.*** *Describe which columns are features (independent variables) and which ones are your target/class/dependent variables.*

Independent:
Work_hours
screen_time_hours
meetings_count breaks_taken
after_hours_work
app_switches
sleep_hours
task_completion
isolation_index fatigue_score
burnout_score
day_type_Weekend

Dependent:
burnout_risk

All numerical variables are continuous and were standardized. Most variables represent behavioral metrics, while indices such as isolation_index and burnout_score are scaled measures representing psychological states.

4. Modeling

4.1. ***Description of Algorithm 1 and its model.*** *Please include details about the parameters. Did you fine-tune the parameters?*

Logistic Regression was used as the baseline classifier for this task. The model was configured with class_weight='balanced', which automatically adjusts weights inversely proportional to class frequency. The dataset becomes approximately balanced after combining Medium and High into a single “At Risk” category. max_iter was tuned from the default of 100 to 1000 after the model failed to converge at lower iteration limits, producing convergence warnings that indicated the solver had not found a stable solution. The default lbfgs solver was retained after testing saga and newton-cg, which yielded no meaningful improvement in accuracy or convergence speed on this dataset.

4.2. ***Description of Algorithm 2 and its model.*** *Please include details about the parameters. Did you fine-tune the parameters?*

Random Forest was selected as the second classifier due to its ability to capture non-linear relationships and feature interactions without requiring feature scaling. n_estimators was tested at values of 50, 100, and 200: accuracy plateaued at 100 trees with no significant gains at higher values, making 100 the most efficient choice. max_depth=None was selected after testing fixed depth limits of 5 and 10, both of which were found to underfit the data by restricting the trees from fully capturing the patterns in the behavioral features. class_weight='balanced' was applied to handle class imbalance consistently with the other classifiers. Notably, Random Forest does not require feature scaling, so it was trained directly on the unscaled feature set unlike Logistic Regression and SVM.

4.3. ***Description of Algorithm 3 and its model.*** *Please include details about the parameters. Did you fine-tune the parameters?*

The Support Vector Machine was configured with kernel='rbf' (Radial Basis Function) after comparing against a linear kernel, which produced lower recall on the At Risk class, suggesting that the decision boundary between classes is not linearly separable in this feature space. The RBF kernel maps the data into a higher-dimensional space, allowing the model to find a more flexible boundary. gamma='scale' was selected over gamma='auto' as it produced more stable and consistent performance across the feature set by accounting for both the number of features and feature variance when computing the kernel coefficient. class_weight='balanced' was applied for consistency with the other classifiers. Because SVM is sensitive to feature scale, it was trained on the standardized feature set produced during preprocessing.

4.4. ***Bonus Modeling*** *Please include details about the parameters. Did you fine-tune the parameters?*

Clustering and Estimation:
Beyond the three primary classification models, we conducted bonus modeling through clustering and estimation tasks to better understand burnout patterns from multiple perspectives. For clustering, we excluded both burnout_risk and burnout_score from the input features in order to avoid target leakage and allow the algorithms to identify naturally occurring behavioral profiles. We implemented three clustering models: K-Means, Agglomerative Clustering, and BIRCH. K-Means was configured with n_clusters=3 to match the three burnout risk levels in our data (low, medium and high). Three clusters were selected to examine whether the underlying behavior patterns roughly align with the three observed burnout risk levels. The n_init parameter was increased from the default in order to make the solution more stable across different random centroid initializations. Agglomerative Clustering was configured with n_clusters=3, again matching the three burnout risk levels, and linkage='ward', which was selected because Ward linkage minimizes intracluster variance and therefore produces compact, behaviorally similar groups. BIRCH was configured with n_clusters=3 and threshold=1.5. This threshold was chosen as a moderate value that allows the algorithm to form compact local subclusters without fragmenting the data excessively before the final clustering step. No exhaustive hyperparameter search was performed for the clustering models; instead, parameters were chosen based on interpretability, stability, and alignment with the project's goal of profiling remote workers.
The clustering results revealed several meaningful employee behavior profiles. In the K-Means solution, one cluster represented a low-burnout weekday profile, characterized by lower work hours, screen time, meetings, app switching, isolation, and fatigue, along with higher sleep, more breaks, and stronger task completion. A second cluster reflected a high-strain profile, with the highest work hours, screen time, meetings, app switching, isolation, fatigue, and burnout score, as well as lower sleep, fewer breaks, and lower task completion. The third K-Means cluster was also relatively low-burnout, but was strongly defined by weekend observations, suggesting that K-Means split the lower-burnout population into a weekday-type and weekend-type pattern. In the Agglomerative solution, one cluster again captured a stable low-burnout group, while a second cluster represented an extreme overload group with the most severe values across meetings, screen time, fatigue, isolation, and burnout score. The third Agglomerative cluster formed a moderate-risk transitional group, showing above-average work hours, screen time, after-hours work, fatigue, and burnout score, but not to the same extreme degree as the overload cluster. In the BIRCH solution, one cluster captured a dense high-strain subgroup with very high workload and fatigue indicators, while a second large cluster represented a broad low-burnout majority group with better sleep, lower fatigue, and higher task completion. The third BIRCH cluster was especially notable for consistently high after-hours work while many other variables remained closer to average, suggesting a more specialized behavioral pathway to moderate burnout risk. Across all three clustering approaches, the most consistent pattern was the emergence of one clearly lower-burnout group and one clearly higher-strain group, while the middle portion of the data was handled differently by each method.

For the estimation task, we modeled burnout using both statistical and machine learning approaches. First, we fit two Ordinary Least Squares (OLS) regression models using burnout_score as a continuous dependent variable. The Work Pattern OLS model included work_hours, screen_time_hours, meetings_count, breaks_taken, after_hours_work, app_switches, and day_type_Weekend. The Well-Being OLS model included sleep_hours, task_completion, isolation_index, fatigue_score, and day_type_Weekend. These two specifications were intentionally designed to compare whether work-intensity variables or well-being variables explain more variation in burnout intensity. Because OLS does not involve many tunable hyperparameters, the primary modeling choice was the selection of predictors rather than parameter fine-tuning. In addition, we fit two machine learning regression models using a numerically encoded version of burnout risk (Low = 0, Medium = 1, High = 2). Linear Regression was used as a simple baseline estimation model with default settings, while Random Forest Regressor was configured with n_estimators=100 and max_depth=None to capture nonlinear relationships and interactions between variables. As with the other bonus models, no exhaustive tuning was conducted. These settings were selected as reasonable, stable defaults for exploratory estimation on this dataset.

5. Evaluation

5.1. ***Baseline****. Please describe your baseline for the evaluation.*

We use a majority class classifier as our primary baseline for evaluation. This model always predicts the most frequent class in the training set. In our binary classification setup, the target was converted into Low versus At Risk, where Medium and High burnout risk were combined into the At Risk category to address the original class imbalance. The resulting class distribution is approximately balanced, with about 50.9% Low and 49.1% At Risk observations. Therefore, a majority-class baseline achieves approximately 51% accuracy. This provides a simple and interpretable benchmark: any effective machine learning model should outperform this baseline, demonstrating that it is learning meaningful behavioral patterns rather than relying on class frequency alone.

We also use Logistic Regression as a secondary benchmark because it is simple, interpretable, and linear. This allows us to compare whether more flexible models, such as Random Forest and SVM, provide meaningful improvements beyond a basic supervised learning model.

5.2. ***Metrics****. Describe the metrics and their results.*

Model performance was evaluated using task-appropriate metrics. For classification, we used accuracy, precision, recall, and F1-score. Accuracy measures the overall proportion of correct predictions, while precision and recall evaluate class-specific performance. F1-score balances precision and recall, making it useful for assessing whether a model performs consistently across both Low and At Risk categories. Macro-averaged metrics were emphasized because they weight both classes equally.

After removing potential leakage variables, specifically `burnout_score` and `fatigue_score`, all three classification models substantially outperformed the majority-class baseline. Logistic Regression achieved approximately 95% accuracy, Random Forest achieved approximately 94% accuracy, and SVM achieved approximately 96% accuracy. SVM performed best overall, with the highest accuracy and macro F1-score, while Logistic Regression remained a strong and interpretable benchmark. Random Forest performed slightly lower but remained competitive. These results indicate that the models are learning meaningful behavioral patterns from the cleaned feature set.

For estimation, OLS models were evaluated using R-squared, Adjusted R-squared, F-statistic, AIC, and BIC, while machine learning regression models were evaluated using mean squared error (MSE). The Work Pattern OLS model explained about 50.6% of the variation in `burnout_score`, while the Well-Being OLS model explained about 98.7%, suggesting that variables such as sleep, isolation, task completion, and fatigue are much more closely associated with burnout intensity than work-pattern variables alone. For machine learning regression, Random Forest Regressor achieved a lower MSE (0.0396) than Linear Regression (0.0818), indicating better predictive performance.

For clustering, we evaluated K-Means, Agglomerative Clustering, and BIRCH using SSE, silhouette score, Davies-Bouldin index, Calinski-Harabasz score, and purity. K-Means produced the most compact clusters, BIRCH achieved the best cluster separation, and Agglomerative Clustering showed the strongest alignment with observed burnout labels. However, the relatively modest silhouette scores suggest that burnout-related behavioral patterns overlap rather than forming completely distinct groups.

6. Discussion

*6.1. **Discussion of the Results**. Please provide a thorough and insightful analysis of the results, integrating them with previous/related work.*

Overall, the results show that burnout risk can be predicted, explained, and profiled using behavioral and well-being data. The classification models performed well after the removal of potentially leaky variables. The original near-perfect accuracy suggested that some variables were too closely tied to the target, so removing `burnout_score` and `fatigue_score` made the results more realistic and defensible. After this correction, model performance stabilized around 94–96% accuracy, with SVM performing best overall. This suggests that burnout risk is not simply random; it is meaningfully associated with observable patterns in work behavior, sleep, isolation, and productivity.

However, it is important to note that this dataset is synthetically generated, which introduces an important limitation. Because the data is artificially constructed, the relationships between variables may be cleaner, more structured, or more deterministic than in real-world settings. As a result, model performance may appear higher and more stable than would be expected with real organizational data, where noise, missing values, measurement error, and unobserved factors are more prevalent. In this sense, the results should be interpreted as demonstrating the theoretical potential of these models rather than their guaranteed real-world performance. This also helps explain why some models, particularly the Well-Being OLS model, achieve extremely high explanatory power, which may reflect built-in relationships in the synthetic data rather than naturally occurring patterns.

The estimation results add another layer of insight. The Work Pattern OLS model explained about half of the variation in burnout intensity, showing that work-related factors such as work hours, screen time, after-hours work, and app switching are relevant. However, the Well-Being OLS model explained substantially more variation, suggesting that recovery and stress-related variables are more directly connected to burnout intensity. While this aligns with prior research on burnout and remote work, the strength of these relationships may be amplified due to the synthetic nature of the data.

The clustering results complement the supervised models by showing that employees can be grouped into different behavioral profiles, but those profiles are not perfectly separable. This is important because burnout likely exists on a spectrum rather than as sharply distinct categories. K-Means, Agglomerative Clustering, and BIRCH each highlighted different aspects of the data: compactness, label alignment, and separation. Together, these models suggest that burnout risk is best understood through multiple perspectives rather than one single modeling approach. At the same time, the relatively modest cluster separation reinforces the idea that even in a structured dataset, behavioral patterns overlap, and this overlap would likely be even more pronounced in real-world data.

*6.2. **Recommendations and implications of your report**. Provide relevant recommendations, develop thoughtful implications, and suggest specific actionable directions for the future.*

Based on the results, organizations should monitor not only workload metrics, but also well-being indicators such as sleep, fatigue, isolation, and task completion. While work hours and screen time are important, the estimation results suggest that recovery and stress-related variables may be stronger indicators of burnout intensity. This implies that effective interventions should go beyond reducing workload and instead focus on improving recovery, promoting healthy work boundaries, and addressing social isolation among remote workers.

A practical application of this analysis would be the development of an early-warning system that flags employees who appear At Risk based on behavioral patterns. Such a system could help managers proactively identify individuals who may benefit from support, such as adjusted workloads, flexible scheduling, or access to wellness resources. However, any deployment of such models should be handled carefully to ensure ethical use. Predictions should be used to support employees rather than penalize them, and transparency around how predictions are generated is critical to maintaining trust.

Importantly, because this dataset is synthetically generated, these recommendations should be viewed as conceptual and exploratory rather than immediately deployable. The relationships observed in the data may be more structured and less noisy than in real-world settings, which means model performance and feature importance may not fully generalize. Before applying these models in practice, organizations should validate them on real employee data, where factors such as reporting bias, missing data, and unobserved variables may affect performance. This validation step is essential to ensure that the models remain reliable, fair, and interpretable in real-world environments.

Future work should extend this analysis by incorporating longitudinal data, which would allow burnout to be studied over time rather than as a static outcome. Additional features such as team dynamics, managerial support, and self-reported well-being could also improve model robustness. Clustering-based profiling could be further developed to identify distinct employee segments, enabling more targeted interventions—for example, employees with high workload may need scheduling support, while employees experiencing isolation may benefit from increased team interaction or structured social engagement.

Overall, while the current models demonstrate the potential to predict and understand burnout, their real value lies in informing data-driven, human-centered workplace policies. By combining predictive modeling with thoughtful organizational practices, companies can move toward more proactive and supportive approaches to employee well-being.

7. Code and Data Availability

Please provide the link to your GitHub repository here:

https://github.com/chuang26-hue/datascience-burnout-project

Link to dataset:
https://www.kaggle.com/datasets/sonalshinde123/work-from-home-employee-burnout-dataset

8. Members' Contributions

You must describe what each member contributed to this report. We will adopt the CRediT Taxonomy to describe each team member's individual contributions. The submitting team member is responsible for providing the contributions of all authors at submission. We expect that all team members will have reviewed, discussed, and agreed to their individual contributions ahead of this time.

Contributor Role	**Role Definition**	**Team Members (List your names here)**
Conceptualization	Ideas; formulation or evolution of overarching research goals and aims.	Isabel, Phoebe, Christian, Daniel
Data Collection	Activities to search for, obtain, download datasets.	Isabel, Phoebe
Data Curation	Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later reuse.	Isabel, Phoebe, Christian, Daniel
Formal Analysis	Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data.	Isabel, Phoebe, Christian, Daniel
Investigation	Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection.	Isabel, Phoebe, Christian, Daniel
Methodology	Development or design of methodology; creation of models	Christian, Daniel
Project Administration	Management and coordination responsibility for the research activity planning and execution.	Isabel, Phoebe
Software	Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code components.	Isabel, Phoebe, Christian, Daniel
Validation	Verification, whether as a part of the activity or separate, of the overall replication/reproducibility of results/experiments and other research outputs.	Isabel, Phoebe
Visualization	Preparation, creation and/or presentation of the published work, specifically visualization/data presentation.	Isabel, Phoebe, Christian
Writing – Original Draft Preparation	Creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation).	Isabel, Phoebe
Writing – Review & Editing	Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre- or post-publication stages.	Isabel, Phoebe

References:

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